

RAG-based AI chatbot for Type 2 Diabetes Health Literacy — Development & Evaluation Study

Background

Unreliable online
diabetes information

LLM + RAG
Interactive, accurate
patient education

Objective
Evaluate a custom
RAG chatbot for T2DM

AI

Fixed prompt +
validated sources

Methods

Evaluation
44 curated questions

Appropriateness

Source match

Evaluation 2
16-query simulated consult

P Typical
patients

Specialist review
Appropriate / Partly / Inappropriate

Source attribution

- Matched reference doc
- Partly matched
- General knowledge
- Unmatched

Results & Conclusions

44
questions
evaluated

73%
cited reference
documents

94%
fully appropriate
(sourced Qs)

Evaluation 1 — source attribution (n=44)

32 — cited sources (73%) 12 (27%)

Of 32 sourced responses — appropriateness

30 fully appropriate (94%) 2

Of 12 general knowledge responses

11 appropriate (92%) 1 inapp.

100%

Simulated consult (16 Qs) — all fully appropriate
& sourced from reference documents

Key takeaways

- ✓ RAG chatbot delivers clinically credible, empathetic responses
- ✓ Source citations enhance transparency and patient trust
- ⓘ General-knowledge gaps signal where reference docs need expansion
- ✓ Patient-centric documents improve RAG response quality
- ✓ Approach is adaptable to other chronic conditions

RAG = Retrieval-Augmented Generation · T2DM = Type 2 Diabetes Mellitus · LLM = Large Language Model

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